

Presenting Smaller Models: Distilling What We Know

A Discussion of the Article by Samuel Greengard

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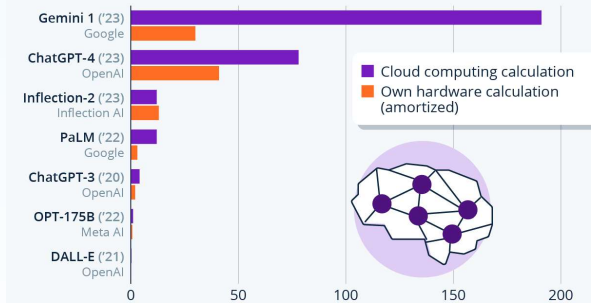


Background

- Powerful models are large!
- Lots of overhead
- Desire to make models smaller without losing performance
- What is SparseGPT?
 - Smaller, Focused, Safer

The Extreme Cost of Training AI Models

Estimated cost of training selected AI models (in million U.S. dollars), by different calculation models

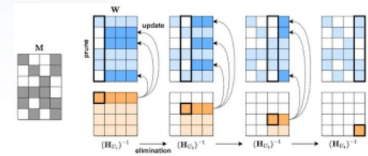


Rounded numbers. Excludes staff salaries that can make up 29 - 49% of final cost (including equity)
Source: Epoch AI



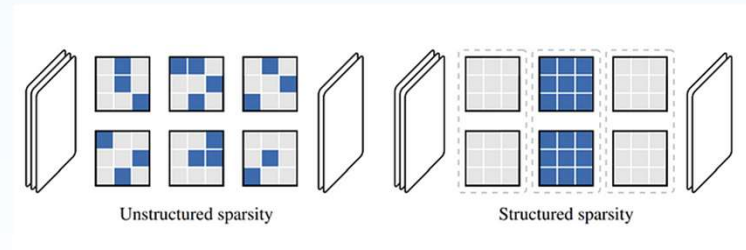
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SPARSE-GPT REMOVE 100 BILLION PARAMETERS



Breaking the Model & Learning the Language of AI

- Objective: Scale down, keep results
- Parameters are just needed for training
- Few useful techniques:
 - Quantization = lowering precision bit amount
 - Sparsity = removing unnecessary values
 - Structured Sparsity = Removing groups of parameters
 - Unstructured Sparsity = Removing redundant parameters
- Teacher and Student networks
- SparseGPT has shown us that different techniques and full application of current techniques are needed



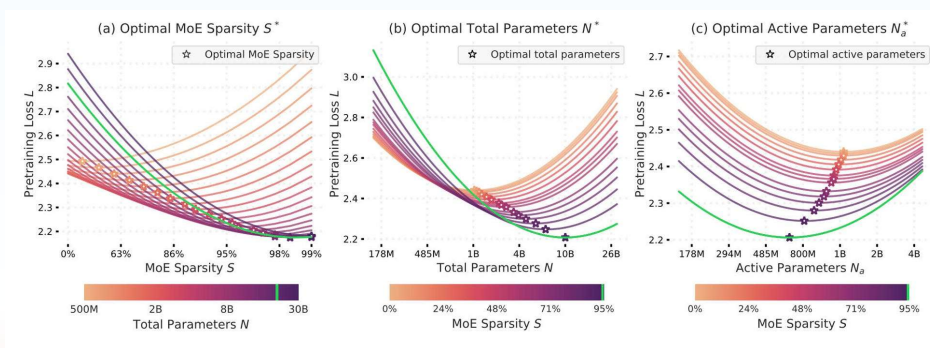
Lowering Noise, Raising Signals, & Outcome Matter

- Implementation of what we know from SparseGPT
- Compactness leads to resistance to noise
- Prior techniques need to be pushed further
- Focus is still the outcome, not just compression
- Mistakes on compression can lead to hallucinations
- Consideration of usage, and !!warnings!!



Personal Opinions

- IT WWWOOOOORRRKKKSSS!
- DeepSeek: Optimizing sparsity and parameters



deepseek



Conclusion & References



- It will be more common for models to use these techniques to become smaller and yet more effective!
- Thank you for watching!
- Primary Source:
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